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202501040044

CODE

TANTRA

PRACTICAL

04

CS1 C13

ESSENTIALS OF DATA SCIENCE

PRACTICE LAB ASSIGNMENT

4.1.1. PANDAS - SERIES CREATION AND MANIPULATION

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

-

- The program should display the mean of even and odd numbers separately.
Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

INPUT

```
import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
series=pd.Series(numbers)
# Grouping by even and odd numbers and calculating the mean
grouped =series.groupby(series%2==0).mean()

# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in
grouped.index]
print("Mean of even and odd numbers:")
print(grouped)
```

OUTPUT

Average time
0.029 s
29.00 ms

Maximum time
0.031 s
31.00 ms

✓ 3 out of 3 shown test case(s) passed
✓ 3 out of 3 hidden test case(s) passed

✓ Test case 1 31 ms Debug

Expected output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:↵

Odd ····5.0↵

Even ····6.0↵

dtype: float64↵

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:↵

Odd ····5.0↵

Even ····6.0↵

dtype: float64↵

✓ Test case 2 28 ms Debug

Expected output

4 4 4 6 6 2 2 2 10 12 14 16

Mean of even and odd numbers:↵

Even ····6.833333↵

dtype: float64↵

Actual output

4 4 4 6 6 2 2 2 10 12 14 16

Mean of even and odd numbers:↵

Even ····6.833333↵

dtype: float64↵

✓ Test case 3 28 ms Debug

Expected output

1 5 7 9 11 13 15 17 111 21 25

Mean of even and odd numbers:↵

Odd ····21.363636↵

dtype: float64↵

Actual output

1 5 7 9 11 13 15 17 111 21 25

Mean of even and odd numbers:↵

Odd ····21.363636↵

dtype: float64↵

4.1.2. DICTIONARY TO DATAFRAME

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user. Display the DataFrame
- after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user. Display the
- DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column.

INPUT

```
import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
}

# Convert the dictionary to a DataFrame
df = pd.DataFrame(data)

# Display the original DataFrame
print("Original DataFrame:")
print(df)
name=input("New name: ")
age=int(input("New age: "))
# Adding a new row
new_data={"Name":name,"Age":age}

df=df.append(new_data,ignore_index=True)
#df=pd.concat( [df,pd.DataFrame( [new_data] )],ignore_index=True
)

# Display the DataFrame after adding a new row
print("After adding a row:\n",df)

# Modifying a row
i=int(input("Index of row to modify: "))
new_age=int(input("New age: "))
df.loc[i,"Age"]=new_age


# Display the DataFrame after modifying a row
print("After modifying a row:")
print(df)

# Deleting a row
i=int(input("Index of row to delete: "))
df=df.drop(df.index[i]).reset_index(drop=True)
# Display the DataFrame after deleting a row
print("After deleting a row:")
print(df)

# Adding a new column
g_input=input("Enter genders separated by space: ")
genders=g_input.split()
df["Gender"]=genders

# Display the DataFrame after adding a new column
print("After adding a new column:")
print(df)

# Modifying a column

# Display the DataFrame after modifying a column
df["Name"]=df["Name"].str.upper()
print("After modifying a column:")
print(df)
```

OUTPUT

Average time

0.115 s

114.50 ms

Maximum time

0.119 s

119.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1119 ms

Debug

Expected output

Actual output

Original DataFrame:

Original DataFrame:

.....Name..Age

.....Name..Age

0...Alice...25

0...Alice...25

1.....Bob...30

1.....Bob...30

2..Charlie...35

2..Charlie...35

New name: Susan

New name: Susan

New age: 29

New age: 29

After adding a row:

After adding a row:

.....Name..Age

.....Name..Age

0...Alice...25

0...Alice...25

1.....Bob...30

1.....Bob...30

2..Charlie...35

2..Charlie...35

3...Susan...29

3...Susan...29

Index of row to modify: 0

Index of row to modify: 0

New age: 27

New age: 27

After modifying a row:

After modifying a row:

.....Name..Age

.....Name..Age

0...Alice...27

0...Alice...27

1.....Bob...30

1.....Bob...30

2..Charlie...35

2..Charlie...35

3...Susan...29

3...Susan...29

Index of row to delete: 3

Index of row to delete: 3

After deleting a row:

After deleting a row:

.....Name..Age

.....Name..Age

0...Alice...27

0...Alice...27

1.....Bob...30

1.....Bob...30

2..Charlie...35

2..Charlie...35

Enter genders separated by space: F M

Enter genders separated by space: F M

After adding a new column:

After adding a new column:

.....Name..Age..Gender

.....Name..Age..Gender

0...Alice...27.....F

0...Alice...27.....F

1.....Bob...30.....M

1.....Bob...30.....M

2..Charlie...35.....M

2..Charlie...35.....M

After modifying a column:

After modifying a column:

.....Name..Age..Gender

.....Name..Age..Gender

0...ALICE...27.....F

0...ALICE...27.....F

1.....BOB...30.....M

1.....BOB...30.....M

2..CHARLIE...35.....M

2..CHARLIE...35.....M

After deleting a column:

After deleting a column:

.....Name..Gender

.....Name..Gender

0...ALICE.....F

0...ALICE.....F

1.....BOB.....M

1.....BOB.....M

2..CHARLIE.....M

2..CHARLIE.....M

4.1.3. STUDENT INFORMATION

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

INPUT

```
import pandas as pd

# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None, names=
["Name", "Age", "Grade"])

# write your code here..
df=pd.DataFrame(data)
print("First five rows:")
print(df.head(5))
print("Average age:",df["Age"].mean().round(2))
print("Students with a grade up to B")
filtered_df=df[df['Grade'].isin(['A', 'B'])]
print(filtered_df)
```

OUTPUT

Average time
0.034 s
34.00 ms

Maximum time
0.040 s
40.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

✓ Test case 1 40 ms Debug

Expected output

Actual output

studentdata.txt	studentdata.txt
First five rows:	First five rows:
· · · Name · · Age · Grade	· · · Name · · Age · Grade
0 · John · 25 · · · · A	0 · John · 25 · · · · A
1 · Alin · 22 · · · · B	1 · Alin · 22 · · · · B
2 · Emma · 24 · · · · A	2 · Emma · 24 · · · · A
3 · Mike · 23 · · · · C	3 · Mike · 23 · · · · C
4 · Sony · 21 · · · · B	4 · Sony · 21 · · · · B
Average age: 23.29	Average age: 23.29
Students with a grade up to B	Students with a grade up to B
· · · Name · · Age · Grade	· · · Name · · Age · Grade
0 · John · 25 · · · · A	0 · John · 25 · · · · A
1 · Alin · 22 · · · · B	1 · Alin · 22 · · · · B
2 · Emma · 24 · · · · A	2 · Emma · 24 · · · · A
4 · Sony · 21 · · · · B	4 · Sony · 21 · · · · B
5 · Amia · 26 · · · · A	5 · Amia · 26 · · · · A

4.2.1. MONTH WITH THE HIGHEST TOTAL SALES

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it. Also, display
- the total sales for the best month.

LAB ASSIGNMENT

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

```

import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

df['Date'] = pd.to_datetime(df['Date'])

# Calculate total sales for each transaction
df['Total_Sales'] = df['Quantity'] * df['Price']

# Extract month and year from Date
df['Month'] = df['Date'].dt.to_period('M') # Creates YYYY-MM
format

# Find the month with the highest total sales
monthly_sales = df.groupby('Month')['Total_Sales'].sum()
best_month = monthly_sales.idxmax()
highest_sales = monthly_sales.max()

print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")

```

Average time
0.013 s
13.00 ms

Maximum time
0.029 s
29.00 ms

✓ 1 out of 1 shown test case(s) passed
✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 29 ms
Debug

Expected output
sales_data.csv
Best month: 2025-01
Total sales: \$1210.00
Actual output
sales_data.csv
Best month: 2025-01
Total sales: \$1210.00

4.2.2. BEST SELLING PRODUCT

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

INPUT

OUTPUT

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles


```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

product_quantity = df.groupby('Product')['Quantity'].sum()

best_product = product_quantity.idxmax()
highest_quantity = product_quantity.max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

Average time 0.010 s 10.33 ms	Maximum time 0.025 s 25.00 ms	✓ 1 out of 1 shown test case(s) passed ✓ 2 out of 2 hidden test case(s) passed
--	--	--

✓ Test case 1 25 ms Debug

Expected output

sales_data.csv

Best selling product: Product A

Total quantity sold: 23

Actual output

sales_data.csv

Best selling product: Product A

Total quantity sold: 23

4.2.3. CITY THAT SOLD THE MOST PRODUCTS

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city. Find the city that
- sold the most products (based on the total quantity sold).

Sample Data:

INPUT

Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles

```
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

city_sales = df.groupby('City')['Quantity'].sum()

best_city = city_sales.idxmax()

# Display the result
print(f"City sold the most products: {best_city}")
```

OUTPUT

Average time
0.010 s
10.33 ms

Maximum time
0.025 s
25.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 25 ms Debug

Expected output

sales_data.csv

City sold the most products: Los Angeles

Actual output

sales_data.csv

City sold the most products: Los Angeles

4.2.4. MOST FREQUENTLY SOLD PRODUCT PAIRS

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date). Output the product pair/s that was sold most frequently.

Sample Data:

Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago 2025-01-05,Product C,10,30,Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B 2025-01-02: Product A, Product C 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
-

INPUT

Product B and Product C: Appears in transactions on 2025-01-04. Most Frequent Product Combinations: Product A and Product B (2 times)

- Product A and Product C (2 times)
-

```
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)

daily_transactions = df.groupby('Date')['Product'].apply(list)
# Initialize a Counter to store the frequency of each pair
pair_counts = Counter()
# 2. Iterate through each date's transaction
for products in daily_transactions:
    # Sort the products to ensure (A, B) is treated the same as (B, A)
    products = sorted(products)

    pairs = list(combinations(products, 2))
    # Update the counter with these pairs
    pair_counts.update(pairs)

if pair_counts:
    max_count = max(pair_counts.values())
    for pair, count in pair_counts.items():
        if count == max_count:
            print(f"{pair[0]} and {pair[1]}: {count} times")
else:
    print("No product pairs found.")
```

OUTPUT

Average time

0.010 s

9.67 ms

Maximum time

0.022 s

22.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

22 ms

Debug

^

Expected output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

Actual output

sales_data.csv

Product A and Product B: 2 times

Product A and Product C: 2 times

4.2.5. TITANIC DATASET ANALYSIS AND DATA CLEANING

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- 1.Display the first 5 rows of the dataset.
- 2.Display the last 5 rows of the dataset.
- 3.Get the shape of the dataset (number of rows and columns).
- 4.Get a summary of the dataset (using .info()).
- 5.Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
- 6.Check for missing values and display the count of missing values for each column.
- 7.Fill missing values in the ‘Age’ column with the median age.
- 8.Fill missing values in the ‘Embarked’ column with the most frequent value (mode()).
- 9.Drop the ‘Cabin’ column due to many missing values.
- 10.Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

Pas sen ger Id	Sur viv ed	Pcl ass	Na me	Sex	Ag e	Sib Sp	Par ch	Tic ket	Far e	Ca bin	Em bar ked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

OUTPUT


```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())
# 2. Display the last 5 rows of the dataset
print(data.tail())
# 3. Get the shape of the dataset
print(data.shape)
# 4. Get a summary of the dataset (info)
print(data.info())
# 5. Get basic statistics of the dataset
print(data.describe())
# 6. Check for missing values
print(data.isnull().sum())
# 7. Fill missing values in the 'Age' column with the median age
data['Age'].fillna(data['Age'].median(), inplace = True)
# 8. Fill missing values in the 'Embarked' column with the mode
data['Embarked'].fillna(data['Embarked'].mode()[0],inplace=True)
# 9. Drop the 'Cabin' column due to many missing values
data.drop(columns='Cabin',inplace = True)
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data["FamilySize"] =data['SibSp']+data['Parch']
```

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Average time 0.069 s 69.00 ms	Maximum time 0.069 s 69.00 ms	1 out of 1 shown test case(s) passed
Test case 1 69 ms Debug		
Expected output		
PassengerId Survived Pclass Fare Cabin Embarked		
0 1 0 3 7.2500 NaN S		
1 2 1 1 71.2833 C85		
2 3 1 3 7.9250 NaN		
3 4 1 1 53.1000 C123		
4 5 0 3 8.0500 NaN		
[5 rows x 12 columns]		
PassengerId Survived Pclass Fare Cabin Embarked		
886 887 0 2 13.00 NaN		
887 888 1 1 30.00 B42		
888 889 0 3 23.45 NaN		
889 890 1 1 30.00 C148		
890 891 0 3 7.75 NaN		
[5 rows x 12 columns]		
(891, 12)		
<class 'pandas.core.frame.DataFrame'>		
RangeIndex: 891 entries, 0 to 890		
Data columns (total 12 columns):		
# Column Non-Null Count Dtype		
0 PassengerId 891 non-null int64		
1 Survived 891 non-null int64		
2 Pclass 891 non-null int64		
3 Name 891 non-null object		
4 Sex 891 non-null object		
5 Age 714 non-null float64		
6 SibSp 891 non-null int64		
7 Parch 891 non-null int64		
8 Ticket 891 non-null object		
9 Fare 891 non-null float64		
10 Cabin 204 non-null object		
11 Embarked 889 non-null object		
dtypes: float64(2), int64(5), object(5)		
memory usage: 66.2+ KB		
None		
PassengerId Survived Pclass SibSp Parch Fare		
count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000		
mean 446.000000 0.383838 2.308642 0.523008 0.381594 32.204208		
std 257.353842 0.486592 0.836071 1.102743 0.806057 49.693429		
min 1.000000 0.000000 1.000000 0.000000 0.000000 0.000000		
25% 223.500000 0.000000 2.000000 0.000000 0.000000 7.910400		
50% 446.000000 0.000000 3.000000 0.000000 0.000000 14.454200		
75% 668.500000 1.000000 3.000000 1.000000 0.000000 31.000000		
max 891.000000 1.000000 3.000000 8.000000 6.000000 512.329200		
[8 rows x 7 columns]		
PassengerId		

Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype:	int64
Actual output	
PassengerId Survived Pclass Fare Cabin Embarked	
0 1 0 3 7.2500 NaN S	
1 2 1 1 71.2833 C85	
2 3 1 3 7.9250 NaN	
3 4 1 1 53.1000 C123	
4 5 0 3 8.0500 NaN	
[5 rows x 12 columns]	
PassengerId Survived Pclass Fare Cabin Embarked	
886 887 0 2 13.00 NaN	
887 888 1 1 30.00 B42	
888 889 0 3 23.45 NaN	
889 890 1 1 30.00 C148	
890 891 0 3 7.75 NaN	
[5 rows x 12 columns]	
(891, 12)	
<class 'pandas.core.frame.DataFrame'>	
RangeIndex: 891 entries, 0 to 890	
Data columns (total 12 columns):	
# Column Non-Null Count Dtype	
0 PassengerId 891 non-null int64	
1 Survived 891 non-null int64	
2 Pclass 891 non-null int64	
3 Name 891 non-null object	
4 Sex 891 non-null object	
5 Age 714 non-null float64	
6 SibSp 891 non-null int64	
7 Parch 891 non-null int64	
8 Ticket 891 non-null object	
9 Fare 891 non-null float64	
10 Cabin 204 non-null object	
11 Embarked 889 non-null object	
dtypes: float64(2), int64(5), object(5)	
memory usage: 66.2+ KB	
None	
PassengerId Survived Pclass SibSp Parch Fare	
count 891.000000 891.000000 891.000000 891.000000 891.000000 891.000000	

INPUT

OUTPUT

4.2.6. TITANIC DATASET ANALYSIS AND DATA CLEANING - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- 1.Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- 2.Convert the 'Sex' column to numeric values (male: 0, female: 1).
- 3.One-hot encode the 'Embarked' column, dropping the first category.
- 4.Get the mean age of passengers.
- 5.Get the median fare of passengers.
- 6.Get the number of passengers by class.
- 7.Get the number of passengers by gender.
- 8.Get the number of passengers by survival status.
- 9.Calculate the survival rate of passengers.
- 10.Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Pas sen ger Id	Sur viv ed	Pcl ass	Na me	Sex	Ag e	Sib Sp	Par ch	Tic ket	Far e	Ca bin	Em bar ked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# Create a new column 'IsAlone' which is 1 if the passenger
is alone

data['IsAlone'] = np.where(data['FamilySize']==0, 1, 0)

# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns=['Embarked'])
# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print(mean_age)
# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print(median_fare)
# 6. Get the number of passengers by class
print(data['Pclass'].value_counts())
# 7. Get the number of passengers by gender
print(data['Sex'].value_counts())
# 8. Get the number of passengers by survival status
print(data['Survived'].value_counts())
# 9. Calculate the overall survival rate
survival_rate = data['Survived'].mean()
print(format(survival_rate))
# 10. Calculate the survival rate by gender
survival_by_gender = data.groupby('Sex')['Survived'].mean()
print(survival_by_gender)
```

OUTPUT

Average time
0.032 s
32.00 ms

Maximum time
0.032 s
32.00 ms

1 out of 1 shown test case(s) passed

Test case 1 32 ms Debug

Expected output

29.69911764705882

14.4542

3...491

1...216

2...184

Name: Pclass, dtype: int64

0...577

1...314

Name: Sex, dtype: int64

0...549

1...342

Name: Survived, dtype: int64

0.3838383838383838

Sex

0...0.188908

1...0.742038

Name: Survived, dtype: float64

Actual output

29.69911764705882

14.4542

3...491

1...216

2...184

2...184

Name: Pclass, dtype: int64

0...577

1...314

Name: Sex, dtype: int64

0...549

1...342

Name: Survived, dtype: int64

0.3838383838383838

Sex

0...0.188908

1...0.742038

Name: Survived, dtype: float64

4.2.7. TITANIC DATASET ANALYSIS AND DATA CLEANING - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- 1.Calculate the survival rate by class.
- 2.Calculate the survival rate by embarkation location (Embarked_S).
- 3.Calculate the survival rate by family size (FamilySize).
- 4.Calculate the survival rate by being alone (IsAlone).
- 5.Get the average fare by passenger class (Pclass).
- 6.Get the average age by passenger class (Pclass).
- 7.Get the average age by survival status (Survived).
- 8.Get the average fare by survival status (Survived).
- 9.Get the number of survivors by class (Pclass).
- 10.Get the number of non-survivors by class (Pclass). The Titanic

dataset contains columns as shown below,

Pas sen ger Id	Sur viv ed	Pcl ass	Na me	Sex	Ag e	Sib Sp	Par ch	Tic ket	Far e	Ca bin	Em bar ked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

OUTPUT

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())
# 2. Calculate the survival rate by embarkation location
(Embarked_S)
print(data.groupby('Embarked_S')['Survived'].mean())
# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())
# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())
# 5. Get the average fare by passenger class
print(data.groupby('Pclass')['Fare'].mean())
# 6. Get the average age by passenger class
print(data.groupby('Pclass')['Age'].mean())
# 7. Get the average age by survival status
print(data.groupby('Survived')['Age'].mean())
# 8. Get the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())
# 9. Get the number of survivors by class
# Filter for Survived=1, then count occurrences of Pclass
print(data[data['Survived'] == 1]['Pclass'].value_counts())
# 10. Get the number of non-survivors by class
# Filter for Survived=0, then count occurrences of Pclass
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

Average time
0.038 s
38.00 ms

Maximum time
0.038 s
38.00 ms

1 out of 1 shown test case(s) passed

Test case 138 msDebug

Expected output

Pclass

1...0.629630

2...0.472826

3...0.242363

Name: Survived, dtype: float64

Embarked_S

0...0.506073

1...0.336957

Name: Survived, dtype: float64

FamilySize

0....0.303538

1....0.552795

2....0.578431

3....0.724138

4....0.200000

5....0.136364

6....0.333333

7....0.000000

10....0.000000

Name: Survived, dtype: float64

IsAlone

0...0.505650

1...0.303538

Name: Survived, dtype: float64

Pclass

1...84.154687

2...20.662183

3...13.675550

Name: Fare, dtype: float64

Pclass

1...38.233441

2...29.877630

3...25.140620

Name: Age, dtype: float64

Survived

0...30.626179

1...28.343690

Name: Age, dtype: float64

Survived

0...22.117887

1...48.395408

Name: Fare, dtype: float64

1...136

3...119

2...87

Name: Pclass, dtype: int64

3...372

2...97

1...80

Name: Pclass, dtype: int64

Actual output

Pclass

1...0.629630

2...0.472826

3...0.242363

Name: Survived, dtype: float64

Embarked_S

0...0.506073

1...0.336957

Name: Survived, dtype: float64

FamilySize

0....0.303538

1....0.552795

2....0.578431

3....0.724138

4....0.200000

5....0.136364

6....0.333333

7....0.000000

10....0.000000

Name: Survived, dtype: float64

IsAlone

0...0.505650

1...0.303538

Name: Survived, dtype: float64

Pclass

1...84.154687

2...20.662183

3...13.675550

Name: Fare, dtype: float64

Pclass

1...38.233441

2...29.877630

3...25.140620

Name: Age, dtype: float64

Survived

0...30.626179

1...28.343690

Name: Age, dtype: float64

Survived

0...22.117887

1...48.395408

Name: Fare, dtype: float64

1...136

3...119

2...87

Name: Pclass, dtype: int64

3...372

2...97

1...80

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

OUTPUT

4.2.8. TITANIC DATASET ANALYSIS AND DATA CLEANING - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- 1.Get the number of survivors by gender (Sex).
- 2.Get the number of non-survivors by gender (Sex).
- 3.Get the number of survivors by embarkation location (Embarked_S).
- 4.Get the number of non-survivors by embarkation location (Embarked_S).
- 5.Calculate the percentage of children (Age < 18) who survived.
- 6.Calculate the percentage of adults (Age >= 18) who survived.
- 7.Get the median age of survivors.
- 8.Get the median age of non-survivors.
- 9.Get the median fare of survivors.
- 10.Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())
# 2. Display the last 5 rows of the dataset
print(data.tail())
# 3. Get the shape of the dataset
print(data.shape)
# 4. Get a summary of the dataset (info)
print(data.info())
# 5. Get basic statistics of the dataset
print(data.describe())
# 6. Check for missing values
print(data.isnull().sum())
# 7. Fill missing values in the 'Age' column with the median age
data['Age'].fillna(data['Age'].median(), inplace=True)
# 8. Fill missing values in the 'Embarked' column with the mode
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
# 9. Drop the 'Cabin' column due to many missing values
data.drop(columns='Cabin', inplace=True)
# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data["FamilySize"] = data['SibSp'] + data['Parch']
```

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

INPUT

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True)

# 1. Get the number of survivors by gender
print(data[data['Survived']==1]['Sex'].value_counts())
# 2. Get the number of non-survivors by gender
print(data[data['Survived']==0]['Sex'].value_counts())
# 3. Get the number of survivors by embarked location
print(data[data['Survived']==1]['Embarked_S'].value_counts())
# 4. Get the number of non-survivors by embarked location
print(data[data['Survived']==0]['Embarked_S'].value_counts())
# 5. Calculate the percentage of children (Age < 18) who
survived
print(data[data['Age']<18]['Survived'].mean())
# 6. Calculate the percentage of adults (Age >= 18) who
survived
print(data[data['Age']>=18]['Survived'].mean())
# 7. Get the median age of survivors
print(data[data['Survived']==1]['Age'].median())
# 8. Get the median age of non-survivors
print(data[data['Survived']==0]['Age'].median())
# 9. Get the median fare of survivors
print(data[data['Survived']==1]['Fare'].median())
# 10. Get the median fare of non-survivors
print(data[data['Survived']==0]['Fare'].median())
```

OUTPUT

Average time
0.035 s
35.00 ms

Maximum time
0.035 s
35.00 ms

1 out of 1 shown test case(s) passed

Test case 1 35 ms Debug

Expected output

female ····233

male ·····109

Name: Sex, dtype: int64

male ·····468

female ····81

Name: Sex, dtype: int64

1 ····217

0 ····125

Name: Embarked_S, dtype: int64

1 ····427

0 ····122

Name: Embarked_S, dtype: int64

0.5398230088495575

0.3810316139767055

28.0

28.0

26.0

10.5

Actual output

female ····233

male ·····109

Name: Sex, dtype: int64

male ·····468

female ····81

Name: Sex, dtype: int64

1 ····217

0 ····125

Name: Embarked_S, dtype: int64

1 ····427

0 ····122

Name: Embarked_S, dtype: int64

0.5398230088495575

0.3810316139767055

28.0

28.0

26.0

10.5